

★ RS004 Week 14 — English Worksheet

Topic: Vertical Patrol + Death Effect (Apply) · **Course:** Platformer Game · **Time:** about 45 minutes

This week enemies get **smarter**. A `patrol_type` lets one enemy walk left–right, another go up–down, and a third **chase** the player. When the player dies, a little **explosion** of particles bursts out.

Keep these words handy: `patrol_type`, `vertical`, `chase`, `particle`, `explosion`.

1 · Predict 🧠

Read each snippet. Before you run it, write what you think will happen.

```
if self.patrol_type == "vertical":
    self.rect.y += self.direction * self.speed
    if self.rect.y > self.start_y + self.range_size:
        self.direction = -1
    if self.rect.y < self.start_y:
        self.direction = 1
```

Which way does this enemy move? When does it turn around?


```
elif self.patrol_type == "chase" and player_rect is not None:
    if self.rect.x < player_rect.x:
        self.rect.x += 1
    elif self.rect.x > player_rect.x:
        self.rect.x -= 1
```

The player is to the *right* of this enemy. Which way does it move?


```
for p in self.particles:
    p["x"] += p["vx"]
    p["y"] += p["vy"]
    p["vy"] += 0.3
    p["life"] -= 1
```

Each particle has its own speed. Why does **vy** keep growing (**+= 0.3**)?

2 · Pick the Patrol 🕸️

Match each `patrol_type` to its behaviour.

1. "horizontal" 2. "vertical" 3. "chase"

- A. moves up and down between two heights
- B. follows the player when nearby
- C. walks left and right between two x positions

1 → __ 2 → __ 3 → __

3 · Spot the Bug 🐛

Each snippet is broken. Rewrite it so it works, then explain why the original was wrong.

Bug A — A vertical enemy should bounce between two heights. Right now it floats up forever and leaves the screen.

```
if self.patrol_type == "vertical":  
    self.rect.y += self.direction * self.speed
```

Hint: there is no turn-around check on the y edges.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — A chasing enemy needs the player's position. Right now it crashes with a `NoneType` error.

```
def update(self):  
    if self.patrol_type == "chase":  
        if self.rect.x < player_rect.x:  
            self.rect.x += 1
```

Hint: `player_rect` must be passed in, with a safe default, and used only when it exists.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — Dead particles should be removed so the explosion fades. Right now they pile up and slow the game down.

```
for exp in explosions:
    exp.update()
    exp.draw(screen)
```

Hint: when an explosion `is_done()`, take it out of the list. Loop over a copy while removing.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Mix the Enemies

Start from your working enemy types.

1. Place one of each kind in your level: horizontal, vertical, chase. Write the three `Enemy(...)` lines.

2. Give each kind a different colour or shape so the player can tell them apart. What did you choose?

3. **Stretch:** position an enemy so it guards a coin. Which coin, and which enemy?

5 · Build & Show 📷

Add all three enemy kinds to your own level, each at least once, plus the death explosion.

When it works, send a **photo or video** showing all three enemies and a death burst, then explain what you did. Use these starters — write 4 to 6 sentences.

First, the `patrol_type` decides ...

My chasing enemy follows the player by ...

I gave it a safe default so that ...

The death explosion works by ...

One tricky moment was ...

If I had more time, I would ...

6 · Record Your Walkthrough 🎥

Take a video showing all three enemy kinds and a death. Teach it to someone new. Try to use these words: **patrol_type**, **vertical**, **chase**, **particle**, **explosion**.

1. Show the horizontal, vertical, and chasing enemies.
2. Walk near the chaser and show it follow you.
3. Die on purpose and show the explosion.
4. Read the chase code out loud and explain how it picks a direction.

Write what you will say in your video. Plan it here first.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.